

Custom Loss Function for Embryo Development Stage Classification

1. Introduction

This report presents the implementation and evaluation of a custom loss function for classifying human embryo development stages using deep learning models. The classification task involves 16 distinct stages of embryo development, characterized by severe class imbalance and inherent ordinal relationships between stages.

2. Enhanced Log Loss (ELLoss)

2.1 Mathematical Formulation

The Enhanced Log Loss combines three complementary components to address the challenges of this classification task:

$$L(\text{logits}, y) = \text{mean}(\text{focal} \times \text{CE} + \beta \times \text{ordinal_penalty})$$

Where:

- **CE** = weighted cross-entropy with class weights w_c
- **focal** = $(1 - p_t)^\gamma$, where p_t is the predicted probability of the true class
- **ordinal_penalty** = $|E[\text{class index}] - y|$, where $E[\text{class index}] = \sum(p_i \times i)$

2.2 Component Analysis

Component 1: Weighted Cross-Entropy - Addresses class imbalance by assigning higher weights to underrepresented classes.

Component 2: Focal Term - Down-weights well-classified examples to focus training on difficult samples.

Component 3: Ordinal Penalty - Penalizes predictions far from the true class in the ordinal sequence, encouraging nearby predictions when errors occur.

2.3 Class Imbalance Handling

Class weights are computed as $w_c = N / (C \times n_c)$, where N is the total number of samples, C is the number of classes, and n_c is the number of samples in class c . Weights are clipped between 0.5 and 20.0 to prevent extreme values while ensuring rare classes receive sufficient emphasis during training.

2.4 Properties of a Good Loss Function

Differentiability: All components (softmax, power operations, weighted sum) have well-defined gradients, enabling gradient-based optimization.

Monotonicity: The loss decreases as the predicted probability of the correct class increases, providing a consistent optimization signal.

3. Experimental Setup

3.1 Models and Training Strategy

Four pre-trained CNN architectures were evaluated: VGG16, VGG19, MobileNet, and Inception V3. All models employed a two-phase training strategy: Phase 1 (3 epochs, $lr=1e-4$) trained only the classification head with frozen backbone, followed by Phase 2 (4 epochs, lr reduced to $1e-5$ or $5e-5$) where the top 2 convolutional blocks were unfrozen for fine-tuning.

3.2 Hyperparameters

Loss function: $\gamma = 1.0$, $\beta = 0.05$

Optimizer: Adam with cosine annealing scheduler

4. Results

4.1 Model Performance Comparison

Test set performance across all four models:

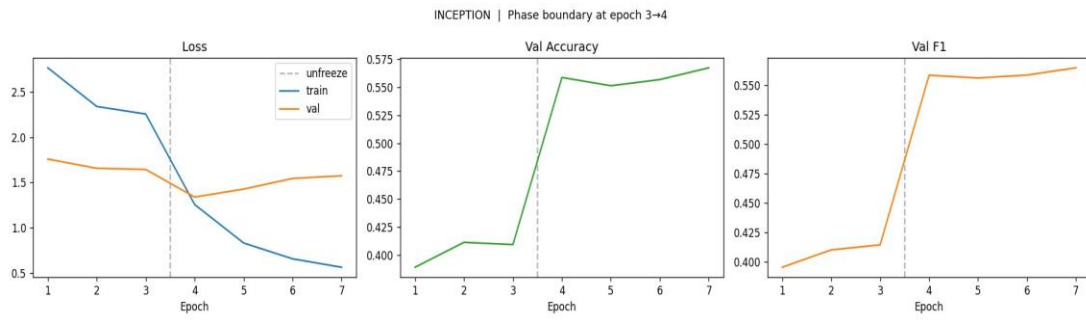
Model	Test Accuracy	Test F1	Best Epoch
Inception V3	0.5789	0.5739	Finetune, Epoch 1
VGG19	0.5728	0.5766	Finetune, Epoch 1
MobileNet	0.5690	0.5767	Finetune, Epoch 1
VGG16	0.5393	0.5606	Finetune, Epoch 1

All models achieved their best performance at the first epoch of the fine-tuning phase, demonstrating the effectiveness of the two-phase training strategy with the custom loss function.

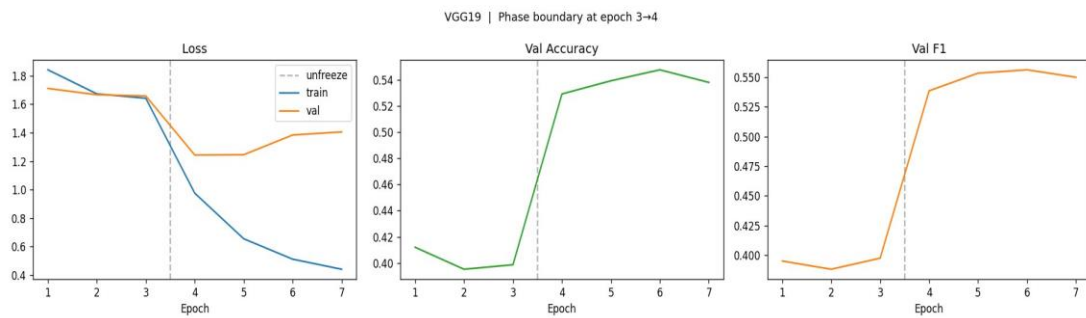
4.2 Training Curves

The following figures show the training progression for each model. The vertical dashed line indicates the phase boundary where the backbone was unfrozen.

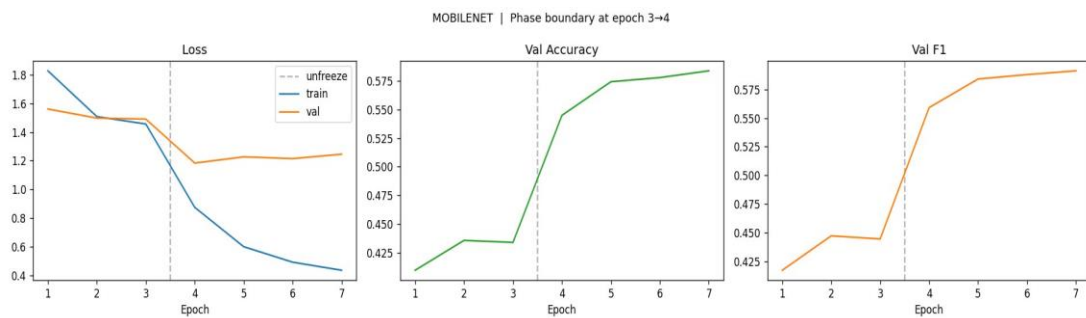
Inception V3



VGG19



MobileNet



VGG16

